Global image phenomics reveals hidden variation and scale-specific environmental geography in thistle capitula

Blinded submission candidate — scientific HOLD retained

Running title

Image phenomics of thistle traits

Abstract

Aim

To test whether public photographs preserve continuous capitulum variation that taxon means hide, and whether the resulting traits show robust, functionally coherent environmental patterns.

Location

Global.

Time period

Observations dated 1985–2026; climate normals for 1981–2010.

Major taxa studied

Source-assigned *Cirsium* and allied Cardueae, used as a replicated-radiation model system.

Methods

An object detector located visible capitula in 46,276 spatially thinned observations from 259 taxa. Deterministic image functions then measured orientation, visible colour, outline and finer geometry under a frozen 27-endpoint contract. Twenty-two endpoints were measured, and five remained explicitly unexecuted. Within- and among-taxon models tested nine predictors in six predeclared blocks. Robustness checks covered global multiplicity, sampling composition, broad spatial structure, residual spatial autocorrelation and 52 historical-placement trees. A complete-18 analysis assessed morphospace and trait organization.

Results

Taxon means compressed 81.4–98.6% of raw visible variation and 28.7–58.5% after equalizing replication. Among 234 endpoint-gradient rows, seven were supported at both scales, 18 within taxa only, three among taxa only, 152 at neither scale and 54 were not comparable. Two among-taxon patterns—lower chroma with higher shortwave radiation and a larger image-referenced orientation angle with higher annual precipitation—passed every sequential gate. The 18-endpoint morphospace remained multidimensional: three principal components explained 42.3%, and within- and among-taxon association geometries were only partly aligned (Spearman = 0.3663).

Main conclusions

The methodological result is that public photographs preserve repeated, coordinate-bearing phenotype distributions that categories and taxon means lose. The ecological result is that

environmental organization depends on trait and scale. The two retained patterns agree with independent evidence for radiation-responsive floral pigmentation and protective downward floral presentation. They are priority candidates for direct tests, not demonstrations of pigment mediation, gravity-referenced orientation, mechanism, selection or adaptation.

Keywords

Cirsium, citizen science, continuous traits, environmental gradients, image phenomics, intraspecific variation, spatial ecology, thistles

1 Introduction

Most global trait datasets give each species one or a few characteristic values. This makes broad comparisons possible, but it hides variation among populations and individuals. Such variation can affect niche breadth, species interactions and coexistence (Messier et al. 2010; Albert et al. 2011; Bolnick et al. 2011; Violle et al. 2012). It can also form a large part of community-level plant trait variation (Siefert et al. 2015). TRY and BIEN have transformed cross-species ecology (Kattge et al. 2011, 2020; Maitner et al. 2018), yet both must rely heavily on taxon summaries and unevenly repeated measurements. We therefore need a complementary way to map the distribution of phenotype itself.

Categories cause a second loss of information. Plant characters are often described as upright or nodding, pale or pigmented, globose or elongate, and strongly or weakly armed. These states are useful for identification, but many underlying phenotypes vary continuously. Numerical traits retain distances within and between categories. They let us ask how much a phenotype varies, where it varies and whether different parts of a structure follow different environmental gradients.

Public biodiversity photographs may provide these repeated measurements. Computer vision already detects and identifies organisms in varied citizen-science images (Van Horn et al. 2018; Wäldchen & Mäder 2018; Rzanny et al. 2019), and multimodal methods are beginning to infer continuous plant traits (Sharma et al. 2026). A visible structure can be represented by an angle, colour coordinate, outline statistic or contour metric. Repeated photographs can then become a geographic sample of visible phenotype, not just a list of photographed species.

Large image collections are not unbiased samples. Observer behaviour, access and taxonomic preference affect which records exist (Troudet et al. 2017; Di Cecco et al. 2021). Light, camera processing, occlusion and viewpoint affect colour, orientation and contour measurements. An unassessable photograph does not show that a trait is absent. Image phenomics must therefore keep trait-specific missingness, measurement provenance and the nesting of heads within photographs. Its measurements are visible image phenotypes, not error-free biological traits.

Biological scale also matters. A trait-environment relationship may occur because one taxon varies across its range, because taxa with different median phenotypes live in different environments, or because both traits and environments follow shared history. These patterns answer different questions. Within-taxon models describe spatial covariation after removing fixed differences among taxa. Among-taxon models describe environmental sorting of taxon medians and need added checks for shared history. A within-taxon gradient is not proof of plasticity, and an among-taxon gradient cannot be read as the same process at a larger scale.

The Asteraceae capitulum is well suited to this test. It is a flower-like structure in which orientation, corolla colour, gross outline and involucral architecture occur together. These components may share some organization, as floral integration theory predicts (Diggle 2014), but they need not share one geographic pattern. We can first measure each component and then ask whether the whole structure reduces to one syndrome.

Thistles provide a strong comparative system. The Carduus-Cirsium group is photographed often and varies widely in capitulum orientation, colour, outline and involucral form. It also contains separate rapid regional radiations rather than one diversification event. Genus-wide phylogenomics identifies rapid Pleistocene radiations after dispersal into Japan and North America (Moreyra et al. 2025). North American *Cirsium* diversified across a continent where precipitation and semiaridity were associated with diversification dynamics (Siniscalchi et al. 2023). The Japanese radiation followed an early-Pleistocene colonization and produced extensive regional endemism. These histories do not show that every regional radiation was adaptive. They do make thistles a useful test of whether categories and taxon means hide repeated phenotypic diversity.

The measured components also have distinct possible functions. Floral orientation can change radiation load, temperature, rain exposure, pollinator presentation and reproductive performance (Chen et al. 2013; van der Kooi et al. 2019; Creux et al. 2021). Summer-blooming Cardueae, including *Cirsium*, can keep capitula cooler than the surrounding air (Herrera 2025). Visible floral colour may respond to temperature, drought, soil chemistry, herbivory and pollinators (Narbona et al. 2018; Vaidya et al. 2018; Borghi et al. 2019; Dalrymple et al. 2020; Sullivan & Koski 2021; Grossenbacher et al. 2026). In *Cirsium palustre*, purple, white and intermediate morphs varied with exposure, population size and year; white flowers sometimes received more pollinator visits (Mogford 1972, 1974a,b, 1978). White-flowered *Cirsium* also occurs in East Asia (Kitamura 1932), but colour alone does not define evolutionary lineages in the Taiwanese *C. japonicum* complex (Chang et al. 2026). Floral displays and scent can attract pollinators and antagonists (Ohashi & Yahara 1998, 2000, 2002; Theis 2006; Theis et al. 2007), and visually defensive structures do not always have their assumed function (Hanley et al. 2007). These studies suggest functions to test; they do not tell us in advance which global associations should appear.

Environmental design needs the same restraint. Temperature and precipitation offer a clear low-dimensional baseline. Radiation, atmospheric drying, wind, growing-season water input and potential productivity may add relevant information. We therefore used predeclared predictor blocks and asked whether they added information beyond the climate core. We did not search an expanding set of rasters after seeing the trait results.

Historical control is important, but history is not the endpoint here. New World *Cirsium* combines high ecological diversity with low nuclear-rDNA divergence (Kelch & Baldwin 2003). Hybridization, polyploidy and incomplete sampling also weaken a single bifurcating account of trait history (Ackerfield et al. 2020; Bureš et al. 2024; Chang et al. 2026). We therefore used available phylogenetic placements only as a robustness check for present among-taxon associations. They cannot identify independent origins, adaptive radiation or convergence.

This study makes two linked contributions. The methodological test asks whether public photographs preserve repeated, spatially explicit continuous phenotypes that categories and taxon means lose. The ecological test asks whether those phenotypes show robust and functionally interpretable environmental geography. We asked four questions: (1) how much visible variation is compressed by one taxon mean; (2) which predeclared gradients align with orientation, colour, outline and finer geometry; (3) at which biological scale does each pattern occur, and does it survive sampling, spatial and historical checks; and (4) do the endpoints form one whole-capitulum syndrome or remain multidimensional?

The endpoint is a bounded description of present phenotype geography. Each trait definition, uncertainty, distribution, association, scale and robustness status remains explicit. Future studies can test function and reconstruct history from these candidates. Claims about independent origins, convergence or adaptation will require nuclear ancestry, explicit origin tests, comparable selection pressures and reproductive-fitness evidence.

2 Methods

2.1 Study design and spatial cohort

We used public biodiversity photographs to quantify continuous capitulum phenotypes across source-assigned taxa of *Cirsium* and allied Cardueae. Source names were retained as operational taxonomic units because the photograph records do not resolve a genus-wide taxonomy. The analysis unit was a coordinate-bearing observation rather than one species-level database row.

The exhaustive source stream contained 406,582 detector-positive observations from 286 taxa. Coordinate filtering retained 392,989 observations from 271 taxa, and a positional-accuracy threshold of 10 km retained 297,293 observations from 259 taxa. Before examining trait values, we retained one observation per taxon by 0.25-degree cell, yielding the frozen spatial cohort of 46,276 observations from 259 taxa. Dates ranged from 1985 to 2026. Missing or unassessable image measurements were never encoded as biological absence or zero (Figure 2; Table 1; Table S2).

2.2 From photographs to a 27-endpoint continuous-trait contract

A single-class YOLO11n detector localized visible capitula at a confidence threshold of 0.25. Detection defined image regions of interest only; downstream traits were calculated with deterministic image functions. The category-free contract registered 27 continuous endpoints across orientation, visible corolla colour, gross outline, floral display, involucral architecture, armature-like projection geometry and validation-only surface-image modules.

Orientation was the signed head-axis angle relative to EXIF-oriented image vertical: 0 degrees upward, 90 degrees horizontal and 180 degrees downward. Image vertical is reproducible but is not a gravity measurement. Colour was represented in visible CIELAB space by lightness, chroma and the sine and cosine of circular hue. Outline and contour variables were dimensionless ratios or normalized geometry. Fine involucral, armature-like and surface endpoints were retained as image phenotypes and were not relabelled as botanical spine length, stiffness, hair density, glands or secretion.

The final artifact-backed run measured 22 of 27 endpoints. Eighteen of 19 inferential endpoints were available; `visible_floret_fraction` and four descriptive colour-composition fractions remained explicit `unexecuted_no_measurement` rows. Hue sine and cosine formed one joint circular inferential unit, producing 26 units from the 27 registered endpoints (Figure 1; Table S1).

2.3 Trait geography and information retained below taxon means

For every endpoint we recorded observation and taxon counts, latitude and longitude limits, and occupancy in 5-degree geographic cells. To quantify information compressed by a one-row-per-taxon database, we decomposed total observation-level sums of squares into variation below and between source-assigned taxon means. Because abundant taxa could dominate the raw partition, we also ran 500 deterministic equal-replication resamples with two observations per eligible taxon. These quantities describe visible image-phenotype variation; they are not heritability, genetic variance or plasticity estimates (Figure 3; Table S3).

2.4 Six predeclared environmental blocks

Nine predictors formed six biological hypothesis blocks. Thermal conditions comprised annual mean temperature and temperature seasonality (CHELSA BIO1 and BIO4). Hydric conditions comprised annual precipitation and precipitation seasonality (BIO12 and BIO15). Radiative and atmospheric exposure comprised mean shortwave radiation and vapour-pressure deficit. Mechanical exposure was represented by mean near-surface wind. Growing-season water input was represented by climatic growing-season precipitation, and resource/productivity by modelled potential net primary productivity. Growing-season precipitation was not taxon-specific flowering-season rainfall, and potential NPP was not measured local resource supply.

All predictors had to retain at least 98% coverage in the phenotype-blind spatial cohort. The nine-variable environmental table was constructed before endpoint-specific missingness filtering; no unavailable variable could be substituted after observing trait associations.

2.5 Within- and among-taxon environmental associations

Within-taxon models used taxon-demeaned standardized traits and predictors, no intercept, and taxon-clustered standard errors. Eligibility required at least 100 observations, ten taxa and two observations per retained taxon. Joint hue used the magnitude and direction of its standardized sine and cosine coefficients under 9,999 permutations restricted within taxa.

Among-taxon models paired taxon trait medians with taxon environmental medians. The primary scope required at least five trait observations per taxon and at least 20 taxa; a separately corrected sensitivity required two observations. Linear endpoints used standardized univariate slopes and 9,999 taxon-label permutations. Joint hue used paired permutations of the sine and cosine slopes.

Benjamini-Hochberg correction formed one global family across all successful unit by predictor rows separately for within taxa, among-taxon minimum-five and among-taxon minimum-two scopes. Historical endpoint tiers did not split multiplicity or remove rows. Each primary minimum-five row was classified as supported at both scales, within taxa only, among taxa only, neither, or not comparable (Figure 4; Tables S4–S6).

2.6 Sampling-composition sensitivity

Every globally FDR-supported row entered a separately frozen retrospective sensitivity; all atlas rows remained reported regardless of entry. Sampling perturbations were selected without using trait outcomes. Within-taxon rows were refitted with equal total weight per taxon. Both scales omitted each of the ten most observed taxa separately and the two most observed taxa jointly, omitted each of six broad geographic regions separately, and retained only observations explicitly classified as native by the frozen WCVP/TDWG join. Linear rows had to retain coefficient sign; circular rows had to remain within 90 degrees of the original coefficient vector. We reported direction and effect-magnitude stability only and calculated no post-selection P or q values (Figure 5; Table S7; Figure S2).

2.7 Broad-space and historical-placement sensitivity

Globally FDR-supported rows were refitted with a second-order spherical-coordinate basis. Passage required a spatial permutation P below 0.05, preserved sign for a linear endpoint and residual eight-nearest-neighbour Moran P of at least 0.05. This was a broad geographic sensitivity, not removal of every spatial process.

Among-taxon rows passing the broad-space gate were then fitted by univariate Pagel-lambda PGLS on 52 audited dated-backbone placement trees: two deterministic scenarios and 50 randomized within-genus placements. A final candidate had to retain direction and P below 0.05 on every tree. Because only 54 historical-atlas taxa were direct backbone tips, this procedure is a placement sensitivity rather than a resolved species-tree correction (Figure 5; Table 2; Table S8; Figures S3-S4).

2.8 Secondary whole-capitulum synthesis

After endpoint-level inference was complete, a secondary complete-18 analysis asked whether the measured traits reduced to one capitulum syndrome. Standardized taxon medians were analysed by principal components, and within- versus among-taxon association matrices were compared by measurement module. This synthesis did not select endpoints or alter the endpoint-level multiplicity families (Table S9).

2.9 Claim boundary

The study is a retrospective exploratory analysis of visible image phenotypes and their present spatial associations. It does not identify plasticity, local adaptation, selection, causal environmental mechanisms, botanical identity of validation-only proxies, independent evolutionary origins or convergence. The two final rows are termed adaptive-pattern candidates under current sequential controls, not demonstrations of adaptation.

3 Results

3.1 Taxon means hid substantial visible variation

The frozen spatial cohort contained 46,276 observations from 259 source-assigned taxa. Twenty-two of 27 registered endpoints were measured, producing 374,255 analysis-eligible observation by endpoint values. Coverage ranged from 909 observations and 85 taxa for validation-only surface proxies to 40,510 observations and 253 taxa for visible CIELAB lightness and chroma. Measured endpoints occupied 190 to 431 five-degree cells (Figures 1–2; Tables S1–S2; Figure S1).

Across the 22 measured endpoints, 0.8137 to 0.9863 of raw observation-level variation occurred below source-assigned taxon means. The equal-replication median ranged from 0.2869 for hue sine to 0.5846 for basal taper ratio. Thus, even after limiting each eligible taxon to two observations, one taxon mean compressed 28.7–58.5% of visible variation, while the unbalanced observed sample compressed 81.4–98.6% (Figure 3; Table S3).

3.2 Trait-environment links depended on trait and scale

Eight predictors had complete coverage and wind had 99.989% coverage. The within-taxon atlas contained 189 executed tests and 45 explicit unexecuted rows; 26 executed rows passed the global FDR family. Supported rows occurred in all six blocks, including orientation with annual mean temperature, chroma with VPD, several involucral geometry endpoints with precipitation metrics, surface-image endpoints with temperature or growing-season precipitation, and joint hue across multiple gradients.

The primary among-taxon atlas contained 180 successful tests, 45 unexecuted rows and nine no-finite-variation rows. Ten rows passed its separate global FDR family. Linear associations comprised a larger orientation angle with annual precipitation (standardized effect 0.3044, $q = 0.0210$), a larger orientation angle with growing-season precipitation (0.2581, $q = 0.0338$), lower chroma with higher shortwave radiation (-0.3454, $q = 0.0060$), and greater bract projection-peak density with VPD (0.4437, $q = 0.0468$). Joint hue was associated with BIO1, BIO4, BIO15, radiation, wind and NPP. Seven rows repeated in the minimum-two sensitivity.

Across the complete 234-row primary scale grid, seven rows were supported at both scales, 18 within taxa only, three among taxa only, 152 at neither scale and 54 were not comparable. The three among-only rows were orientation with annual precipitation, orientation with growing-season precipitation and chroma with radiation. Most supported involucral and surface-proxy rows were within-taxon only (Figure 4; Tables S4–S6).

3.3 Sampling was global but strongly uneven

Europe contributed 23,337 observations (50.43%) and North America 19,357 (41.83%); together they contributed 92.26%. The Northern Hemisphere contributed 94.91%. *Cirsium vulgare* contributed 28.54%, *C. arvense* 25.49% and the ten most observed taxa 79.93%. The taxon-level median was six observations; 51 of 259 taxa had one observation and 150 had fewer than ten.

All 674 declared sampling scenarios were evaluable. All ten selected among-taxon pairs retained direction in every applicable scenario. Sixteen of 26 selected within-taxon pairs retained direction throughout; ten reversed in at least one scenario. Four of the five within-taxon broad-space passes were also directionally stable throughout. Projection roughness with radiation reversed under equal taxon weighting and after jointly omitting the two dominant taxa (Table S7; Figure S2).

3.4 Only two among-taxon patterns passed every gate

Five within-taxon rows passed the broad-space gate: orientation-BIO1, chroma-VPD, involucre length/width-BIO15, projection roughness-radiation and surface high-frequency energy-BIO1. Their spatial standardized effects were 0.0265, -0.0421, 0.0602, -0.1392 and 0.1235, respectively. These rows describe within-taxon spatial covariation and were not treated as plasticity or local adaptation.

Two among-taxon rows passed the broad-space gate. Chroma decreased with radiation (spatial beta = -0.7124, spatial P = 0.001, residual Moran P = 0.217), and orientation angle increased with annual precipitation (spatial beta = 0.2861, spatial P = 0.017, residual Moran P = 0.082). Both retained direction in all 18 applicable sampling scenarios; their minimum effect-magnitude ratios were 0.615 and 0.781.

Both rows retained direction and P below 0.05 on all 52 historical-placement trees. Chroma-radiation PGLS beta was -0.3454 with P = 0.000024 and Pagel lambda = 0 throughout. Orientation-precipitation PGLS beta ranged from 0.3044 to 0.3045, P from 0.000201 to 0.000231 and lambda from 0 to 0.0537. Under the current sequential controls, these are adaptive-pattern candidates. They do not show an adaptive mechanism or independent origin (Figure 5; Table 2; Table S8; Figure S4).

3.5 Capitulum traits remained multidimensional

The expanded 127-taxon by 18-endpoint morphospace remained multidimensional: the first three principal components explained 42.3% cumulatively. Measurement-module contrasts were modest, and the correspondence between within- and among-taxon association geometry was partial (Spearman = 0.3663). Whole-capitulum structure therefore summarized, but did not replace, the decomposed endpoint results (Table S9).

4 Discussion

4.1 Public photographs preserve variation that taxon means lose

The first contribution is methodological. Public photographs gave repeated numerical phenotypes with coordinates and endpoint-specific missingness. One category or taxon mean cannot preserve this information. In the raw data, 81.4–98.6% of visible variation occurred below taxon means. Even after equalizing replication, the range was 28.7–58.5%. A species-mean database loses this distribution even when its mean is accurate.

The workflow also keeps the measurement connected to its limits. Each endpoint has a declared image definition, assessability rule, uncertainty, geographic support and biological scale. Within-taxon gradients can therefore be separated from turnover among taxon medians. Unsupported and unassessable rows remain visible, and missing measurements are not treated as trait absence. This audit trail is as important as the automated measurement itself.

4.2 Capitulum traits do not form one climate syndrome

The 234-row grid did not support one scale-invariant response of the whole capitulum. Most supported involucral and surface-image associations appeared only within taxa. Orientation-precipitation, orientation-growing-season precipitation and chroma-radiation appeared only among taxa. Seven rows were supported at both scales, but most involved joint visible hue, and several retained residual spatial structure. A coefficient at one scale therefore cannot be assigned to another scale.

The complete-18 synthesis gave the same message. Three principal components explained only 42.3% of taxon-level variation, and within- and among-taxon association geometries were only partly aligned. Whole-capitulum summaries can describe partial organization. They should not replace the endpoint-level geography.

This distinction matters in thistles because Japan and North America contain separate rapid regional radiations. Later studies can ask whether the same trait axes diversified in each radiation or whether different trait combinations entered similar environments. The present data do not answer those historical questions and do not show adaptive radiation. They supply continuous phenotype axes and explicit environmental candidates for those tests.

4.3 Two robust patterns match independent functional ecology

Taxa from higher-radiation environments had lower photographed CIELAB corolla chroma. Independent studies make this direction biologically plausible. Supplemental UV-B promoted anthocyanin biosynthesis in rose petals, and *Clarkia unguiculata* populations from areas with higher solar UV had higher estimated floral anthocyanin concentrations (Hennayake et al. 2006;

Peach et al. 2020). In black dahlia, increasing concentrations of specific anthocyanins eventually reduced CIELAB C*, although the relation depended on pigment mixture and pH (Deguchi et al. 2016). Calibrated digital colour indices can also predict measured anthocyanin in some reproductive tissues (del Valle et al. 2018).

These studies suggest a clear test: lower visible chroma may accompany more anthocyanin in some pigment systems, and radiation may promote that pigmentation. Our data do not yet perform that test. The present chroma value is an uncalibrated colour-space distance across taxa. It is not an anthocyanin assay or a UV-reflectance measurement, so the negative coefficient cannot be converted directly into pigment concentration. Calibrated visible and UV reflectance, petal chemistry or pathway expression, and common-garden radiation experiments are needed.

Taxa from wetter environments also had a larger angle on the frozen image scale, where 0 is upward, 90 is horizontal and 180 is downward. The direction is consistent with more downward floral presentation. Along an Andean precipitation gradient, down-facing flowers became more common in wetter environments regardless of pollination mode (Aizen 2003). In the nodding Asteraceae *Cremanthodium campanulatum*, water and UV-B exposure reduced pollen viability. Artificially erect capitula also produced fewer achenes than naturally nodding capitula, without a matching pollinator preference (Chen et al. 2013). Downward presentation may therefore protect reproductive tissue from rain and radiation.

That mechanism remains a hypothesis. EXIF image vertical is not gravity, and we did not measure rain interception, pollen wetting, viability or fitness. A direct test needs gravity-referenced three-dimensional orientation, flowering-period rain data and experimental angle manipulation.

Both patterns survived the full declared sequence: global multiplicity correction, every tested sampling-composition direction, broad-space refitting, residual spatial diagnostics and all 52 historical-placement trees. This agreement between independent functional evidence and internal robustness makes the patterns strong candidates for direct testing. The gates reduce several specific alternative explanations. They do not establish pigment mediation, gravity-referenced orientation, heritability, selection, fitness benefit, causal environmental response or independent evolutionary origin.

4.4 Sampling and measurement still limit inference

The dataset covers much of the world, but it is not a probability sample. Europe and North America supplied more than 92% of observations. Two taxa supplied 54%, and many taxa had few observations. The sampling audit shows how results changed under declared perturbations. It does not make the sample globally representative or create a second confirmatory analysis.

Photography adds further limits. Camera processing changes visible colour, viewpoint changes two-dimensional outline, and image vertical does not measure gravity. Fine involucral and surface variables still need botanical calibration. Five registered endpoints were not executed, including visible floret fraction. Independent detector boxes, reference measurements, developmental-stage labels, paired flower-background colour controls and repeat-photo remeasurement remain open validation requirements.

4.5 Next tests

The next step is validation, not a wider environmental search. The colour candidate needs calibrated pigment, reflectance and background controls. The orientation candidate needs gravity-referenced angles and direct rain-protection measurements. Both need developmental-stage control and independent repeat measurements.

Historical questions come after these functional tests. Nuclear ancestry data can then be used to ask when supported states arose and whether similar states reflect independent origins, retention, ancestral polymorphism or introgression. Convergence and adaptive convergence require evidence for independent origins, comparable selection pressures and reproductive-fitness effects. Present spatial association cannot supply those claims.

4.6 What the study adds

This study delivers two results. Methodologically, public photographs retain a geographic distribution of visible phenotype that categories and taxon means remove. Ecologically, thistle capitula show multidimensional and scale-dependent environmental organization. Lower chroma under higher radiation and a larger image-referenced orientation angle under higher precipitation are the two strongest current candidates. They now define precise botanical, physiological and evolutionary tests; they do not settle those mechanisms or histories.

5 Conclusion and declarations

The methodological contribution is an auditable route from public photographs to repeated, coordinate-bearing continuous phenotypes: starting from 27 registered endpoints, the study recovered 22 measured capitulum phenotypes across 46,276 observations and 259 source-assigned taxa. Taxon means compressed 81.4-98.6% of raw visible variation and 28.7-58.5% after equalizing replication. The ecological contribution is evidence that environmental organization is endpoint- and scale-specific rather than one capitulum syndrome. Two among-taxon directions—lower visible chroma with higher shortwave radiation and a larger image-referenced orientation angle with higher annual precipitation—are consistent with independent evidence for radiation-responsive floral pigmentation and protective downward floral presentation. Both retained support through the current multiplicity, sampling-composition, broad-space, residual-spatial and historical-placement gates.

The convergence of functional plausibility and robustness makes these priority adaptive-pattern candidates under current sequential controls, not demonstrations of adaptation or mechanism. Camera-recorded chroma cannot be inverted into anthocyanin concentration, and image vertical cannot establish gravity-referenced nodding or rain protection. Citizen-science sampling is uneven, and independent detector, trait-measurement and botanical validation remain open. No claim is made about plasticity, heritability, selection, causal environmental response, independent origins or convergence.

Declarations

Ethics approval was not required for analysis of publicly available biodiversity observations. Source photographs will not be redistributed beyond their individual licence terms. Author contributions, funding, acknowledgements, conflicts of interest and corresponding-author details must be completed by the authors in the journal submission system and separate title page. The current repository package remains a submission candidate on scientific HOLD until the external validation gates listed in `EXTERNAL_COMPLETION_GATES.md` are closed.

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Data and code availability

A versioned submission-candidate bundle accompanies this repository. It contains the blinded Main and Supporting Information files, the formatted Main Tables 1–2 and Tables S1–S10 workbook, derived CSV tables, model outputs, figure sources and SHA-256 reproducibility manifests.

The public GitHub repository is not an anonymous review link. Before submission, the exact bundle will be copied to an anonymized read-only location without Git history or account identifiers. Source photographs will not be redistributed beyond their individual licence terms. GitHub Actions artifacts are temporary execution records, not a durable archive.

After the independent validation gates, author metadata and repository licence are complete, the exact supporting version will be deposited in a durable repository with a persistent digital object identifier. This statement will then give the final repository name, DOI and access terms.

Tables

Table 1. Compact analytical roadmap for the global public-image study of thistle capitula. Each stage identifies its frozen scope, operation or decision gate, and linked display; the full eight-column roadmap is Table S10.

analytical stage	frozen scope	operation and gate	linked display
Photo localization	Visible-capitulum bounding boxes	YOLO localization only; deterministic functions score traits after cropping. Gate: A box defines the image unit; independent detector validation remains an external gate	Figure 1; Tables S1-S2
Cohort and realized domain	46,276 observations; 259 source-assigned taxa	Maximum 10-km positional uncertainty; one observation per taxon by 0.25-degree cell. Gate: Freeze the realized sampling domain before examining endpoint values	Figure 2; Table S2
Continuous measurement contract	27 registered endpoints; 22 measured; five explicit unexecuted	Deterministic orientation, visible-colour, outline and fine-geometry measurements. Gate: Missing or unassessable measurement remains missing evidence	Figure 1; Table S1
Taxon-mean information loss	22 measured endpoints	Raw sums-of-squares partition plus 500 two-observations-per-taxon resamples. Gate: Describe visible image-phenotype information loss, not genetic variance	Figure 3; Table S3
Within/among environmental atlas	26 inferential units by nine predictors in six blocks; 234 canonical rows	Taxon-demeaned clustered models and taxon-label permutations; separate global BH families. Gate: Classify every row as both, within-only, among-only, neither or not-comparable	Figure 4; Tables S4-S6
Sampling-composition audit	674 evaluable retrospective scenarios	Direction and effect-magnitude retention only; no post-selection P values. Gate: Annotate sensitivity without converting the sample into a probability sample	Figure 5; Table S7
Broad-space gate	26 within-taxon and ten among-taxon rows	Spatial-basis refit, permutation P and eight-nearest-neighbour residual Moran diagnostic. Gate: Five within and two among rows pass the declared broad-space screen	Figure 5; Table S8
Historical-placement gate	Two deterministic and 50 randomized within-genus placement trees	Pagel-lambda PGLS sensitivity on 52 trees per row. Gate: Two adaptive-pattern candidates under current controls; not adaptation, mechanism or convergence proof	Figure 5; Table S8

Table 2. Frozen evidence chain for the two among-taxon candidate patterns in the global thistle-capitulum cohort. Beta denotes a standardized coefficient; BH denotes Benjamini-Hochberg; P values are descriptive outputs of the predeclared gates. Stability under these controls is not evidence of adaptation, convergence or mechanism.

candidate pattern	global among taxon	sampling composition	broad space gate	historical placement
Lower camera-recorded corolla chroma with higher shortwave radiation	n = 143; beta = -0.345; global BH q = 0.006	all declared directions stable; minimum effect-magnitude ratio = 0.615	beta = -0.712; permutation P = 0.001; residual Moran P = 0.217	P < 0.05 on 52/52 trees; P range 2.39e-05-2.39e-05; lambda range 0.000-0.000
Larger image-referenced orientation angle with higher annual precipitation	n = 142; beta = +0.304; global BH q = 0.021	all declared directions stable; minimum effect-magnitude ratio = 0.781	beta = +0.286; permutation P = 0.017; residual Moran P = 0.082	P < 0.05 on 52/52 trees; P range 2.01e-04-2.31e-04; lambda range 0.000-0.054

Figures

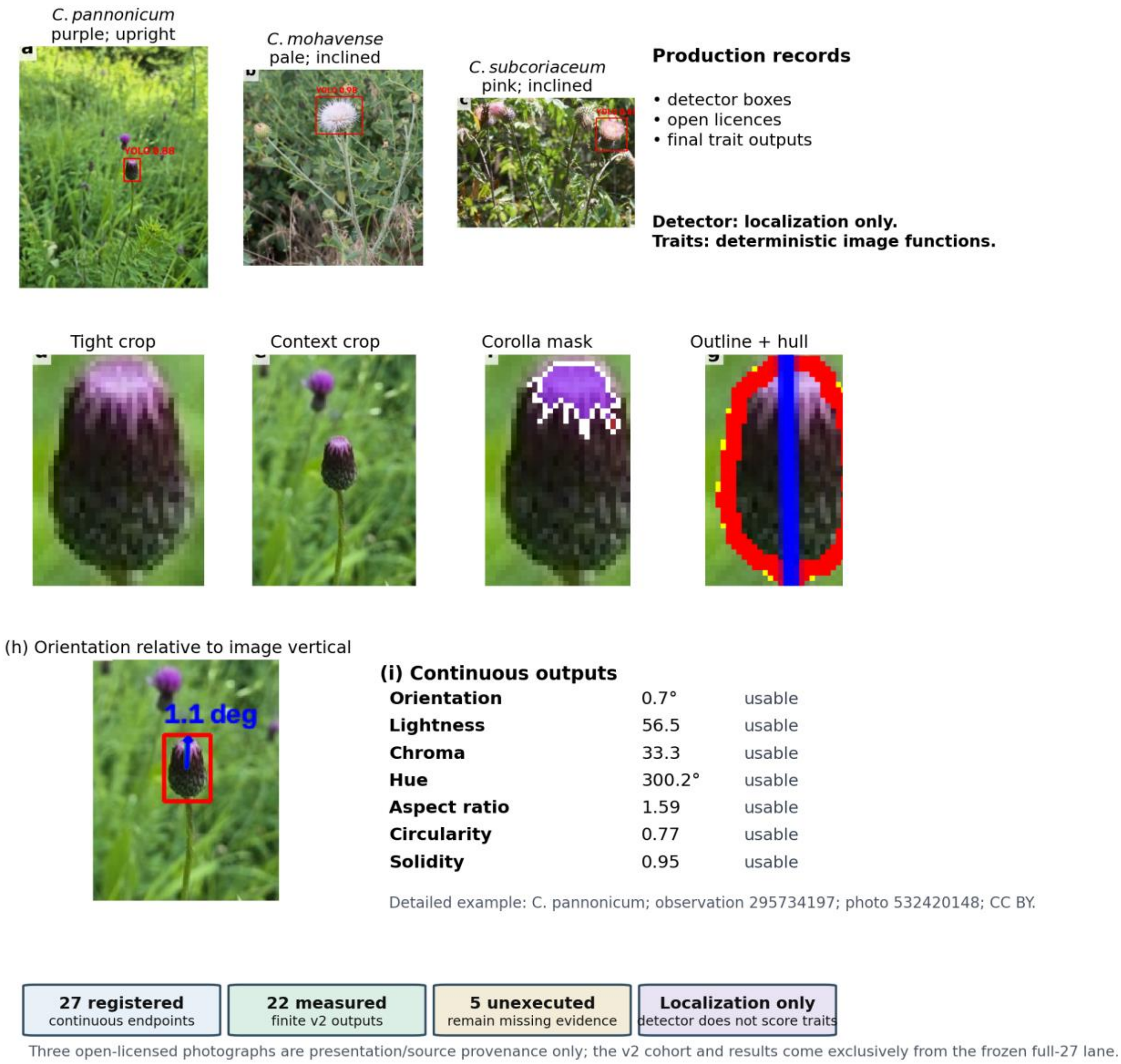

Figure 1. From public photographs to deterministic continuous measurements in the global Carduus-Cirsium thistle-capitulum study. Three open-licensed photographs and their detector/crop examples are presentation/source provenance only; all numerical result statements are current v2.

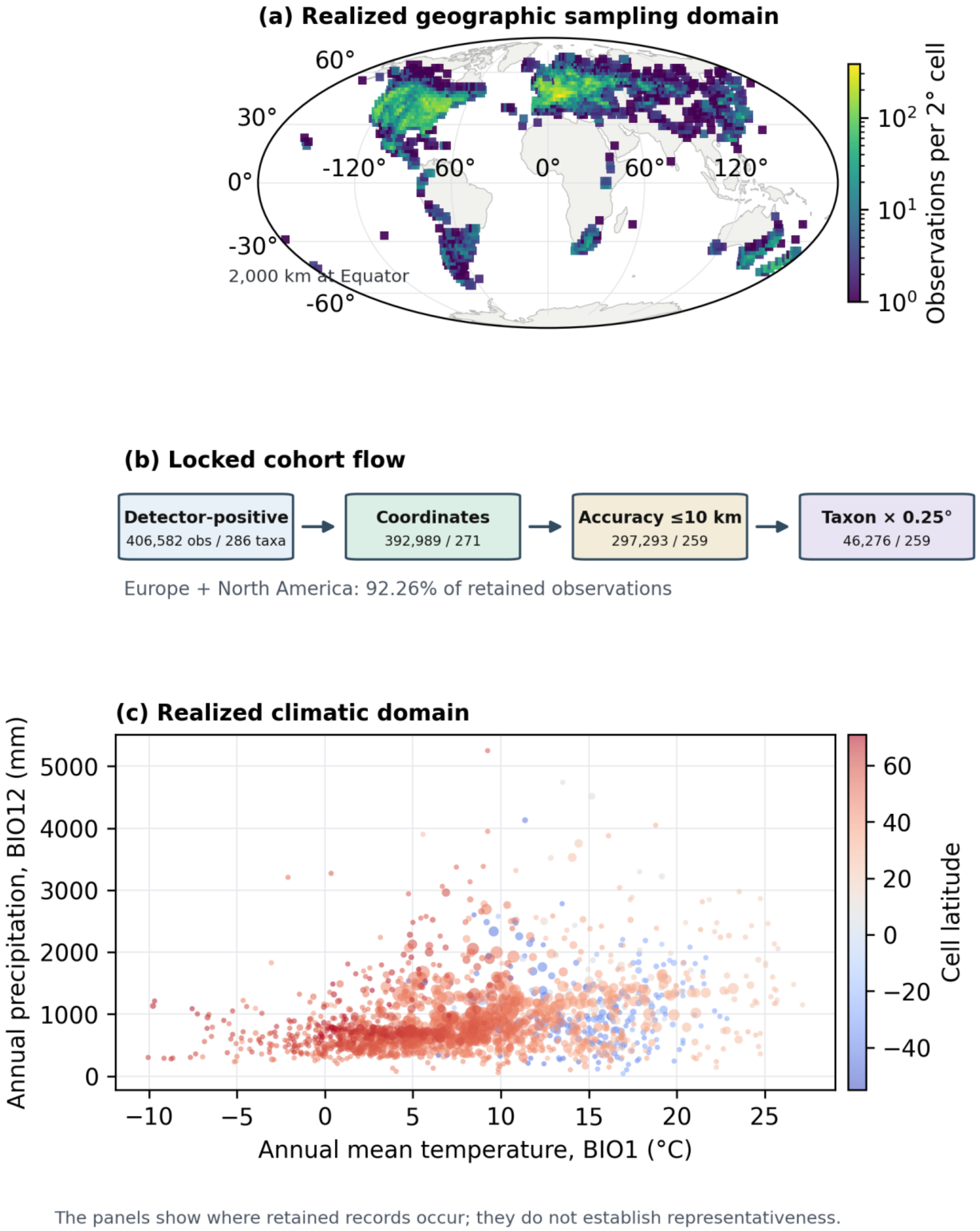

Figure 2. Realized global sampling domain of the frozen public-image cohort of thistle capitula. (a) Disclosure-safe observation density in a Mollweide equal-area projection with a 2,000-km equatorial scale, (b) locked filtering flow to 46,276 observations and 259 source-assigned taxa, and (c) the realized annual-temperature (BIO1) and annual-precipitation (BIO12) domain. The panels do not establish representativeness.

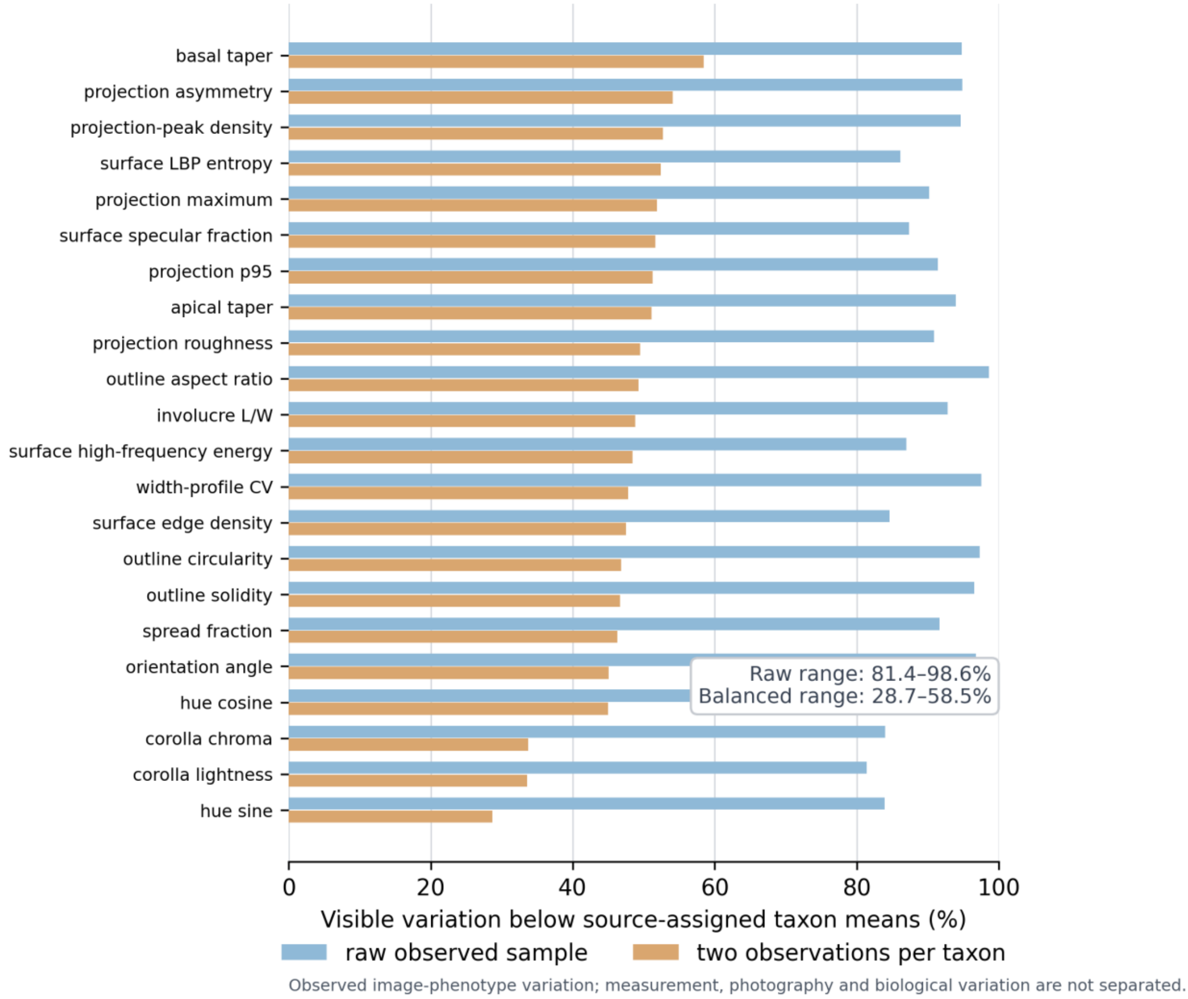

Figure 3. Visible image-phenotype variation compressed by source-assigned taxon means across the global public-image cohort of thistle capitula. Horizontal bars compare the raw observed sample with deterministic equal-replication resamples across 22 measured endpoints; the partition is not genetic variance.

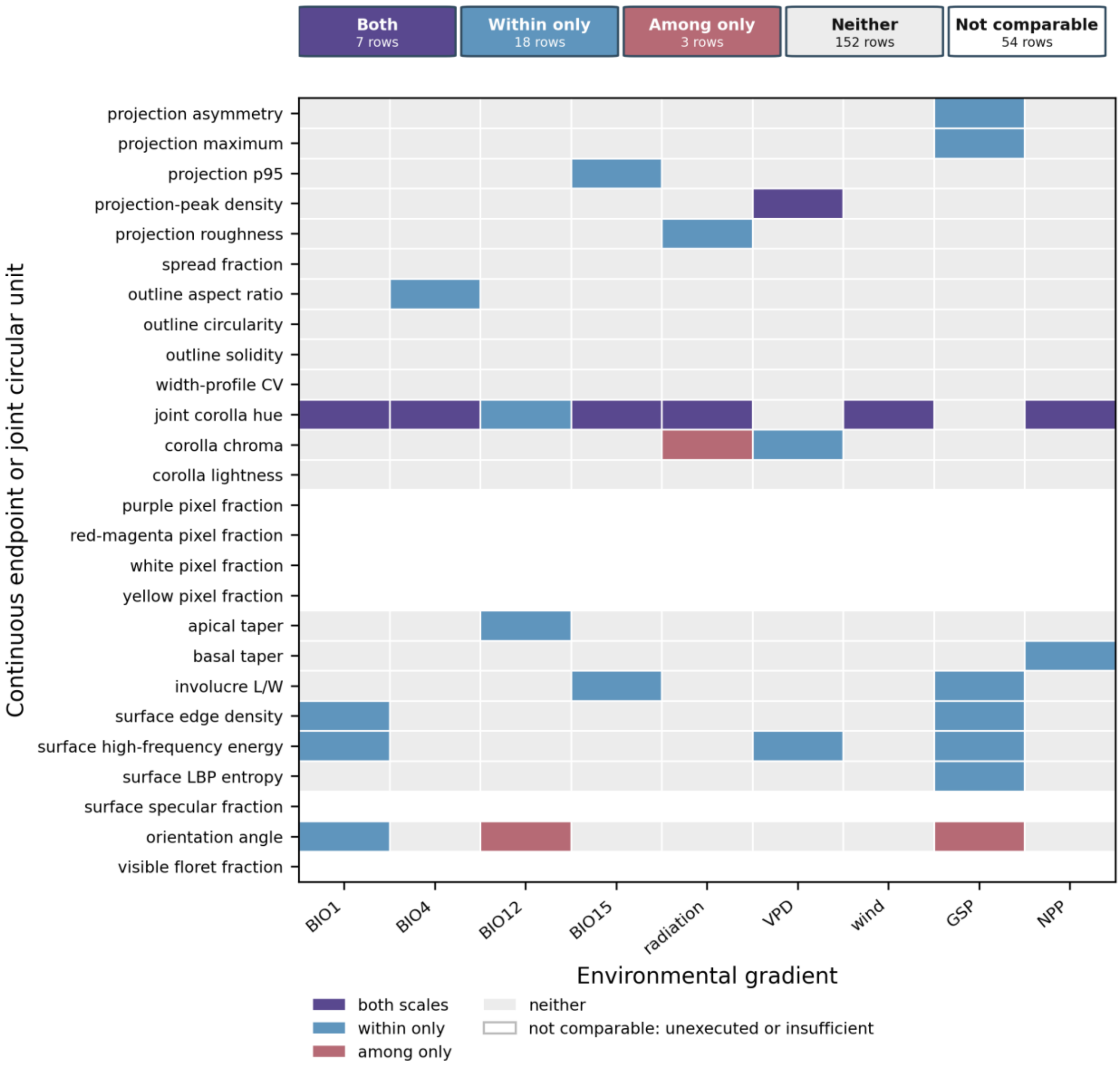

Figure 4. Endpoint-specific environmental alignment within and among source-assigned thistle taxa across the global public-image cohort. The complete 234-row heat map distinguishes both-scale, within-only, among-only, neither and not-comparable rows; joint hue is one circular inferential unit.

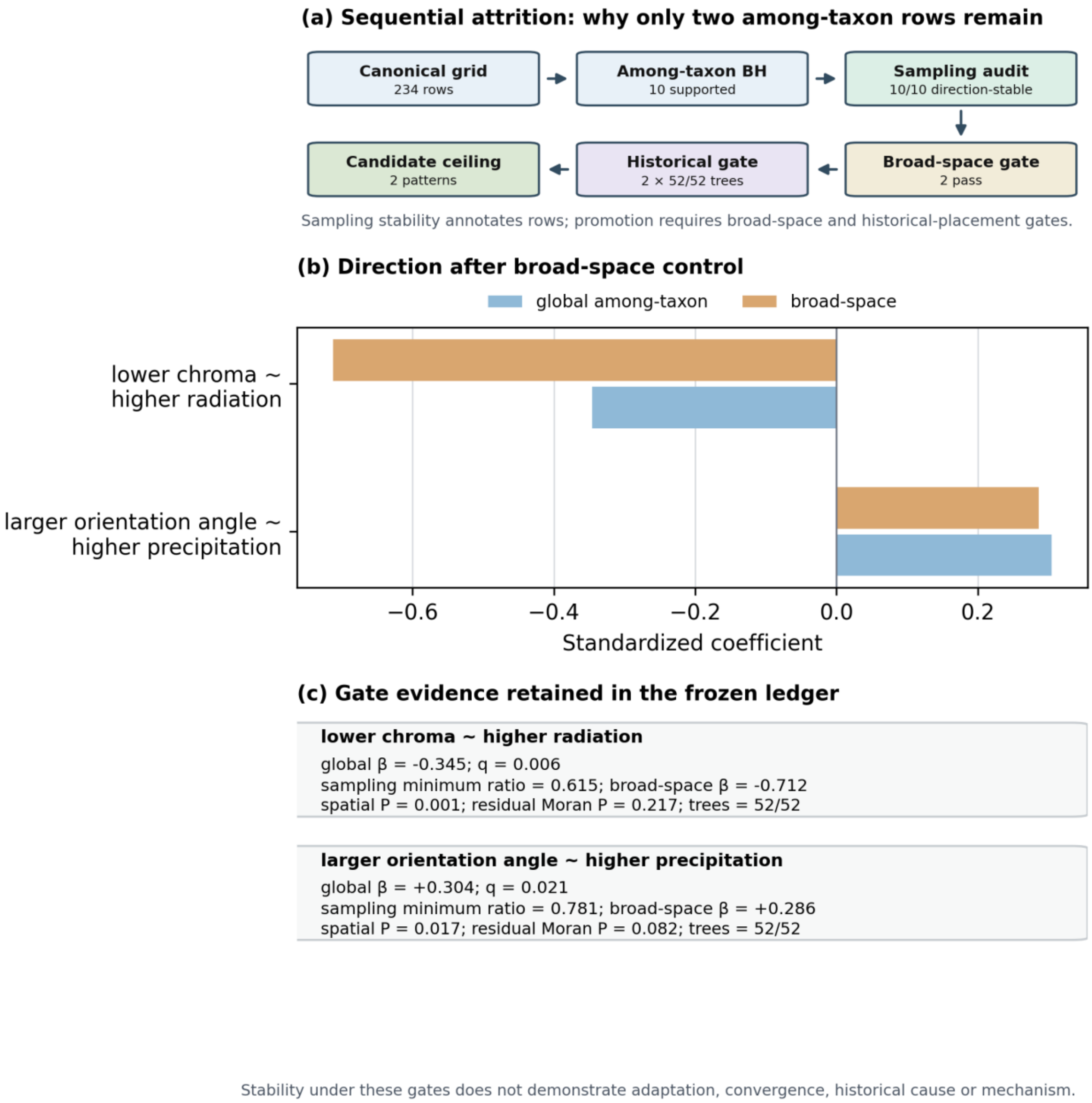

Figure 5. Sequential attrition and robustness of the two final candidate patterns in the global thistle-capitulum cohort. (a) The 234-row grid is reduced through among-taxon multiplicity, sampling annotation, broad-space and 52-tree placement gates; (b-c) retain the coefficient and gate evidence for chroma-radiation and image-referenced orientation-precipitation. Stability is not evidence of adaptation, convergence or mechanism.